

Thursday 6:44 AM

A magnetic field $\mathbf{B}$ has flux surfaces if there exists a single-valued function of position, ψ , such that $\mathbf{B} \cdot \nabla \psi = 0$. We are interested in the case of nested toroidal flux surfaces, where surfaces of constant ψ in some domain are closed, bounded, and topologically toroidal, except for the magnetic axis, where the set of constant ψ is a closed curve inside the toroidal surfaces. Your task is to find an example of a non-axisymmetric magnetic field in a toroidal subset of $\mathbb{R}^3$ with nested toroidal flux surfaces such that ψ is non-constant and is a smooth function of position. ψ should be non-negative, with the set $\psi=0$ being the magnetic axis, corresponding to a closed curve. The constant- ψ surfaces for ψ in $(0, \delta]$ for some $\delta > 0$ should be bounded, closed, and topologically toroidal. The divergence of $\mathbf{B}$ must be 0. I specifically want an example in which the rotational transform is nonzero, so if a streamline of $\mathbf{B}$ (a field line) slightly away from the magnetic axis is followed around once toroidally, it comes back to a different point in the (R, Z) plane. It would be even better (though not critical) if i is not an integer so the return map is not the identity. You don't need to prove conclusively that i is nonzero, but at least ensure there are no symmetries of the field that would guarantee i to be 0. Rotational transform can be achieved by giving net (integrated) torsion to the magnetic axis, or by making the elongation of the constant- ψ surfaces in the (R, Z) plane rotate poloidally as the toroidal angle ϕ increases, or through a current density $\mathbf{J} = \text{curl } \mathbf{B}$ along the magnetic axis. Moreover, I would like the magnetic field to be in MHD force balance, so $(\text{curl } \mathbf{B}) \times \mathbf{B} = \nabla p$ for some $p(\psi)$. This is an open problem, so you will need to be persistent and creative to find a solution that hasn't been found by people before. But you have already solved many open problems in mathematics, so I know that with enough perseverance, you can do it!

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I found an explicit, exact family with nonconstant scalar pressure, a nonplanar magnetic axis, nested analytic flux surfaces, and rotational transform $i = -2$.

There is one important qualification: **its one-toroidal-turn return map is the identity**. Thus it satisfies your allowance that noninteger transform is “not critical,” but it does not satisfy the stronger statement that a field line must return to a different point after one toroidal turn. The nonzero transform here is actual winding in the physical (R, Z) cross-section, not merely an angle relabeling.

Below I give the field as a function of Cartesian position and verify its properties. The construction uses the ordinary Euclidean curl and gradient, with no external force or pressure anisotropy. I have not established whether this particular family has appeared previously in the literature.

1. The explicit example

Use dimensionless coordinates and the normalization in your question, so that

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = \nabla p.$$

Choose a parameter

$$0 < \epsilon < 1, \quad a = \sqrt{1 + \epsilon}, \quad b = \sqrt{1 - \epsilon}.$$

Define

$$X = \frac{x}{a}, \quad Y = \frac{y}{b}, \quad s = X^2 + Y^2, \quad F = \sqrt{2s - s^2 - 4z^2},$$

taking the positive square root. Initially work on the open set

$$\mathcal{U}_\epsilon = \{(x, y, z) : 2s - s^2 - 4z^2 > 0\}.$$

In this set $s > 0$, so the following field is real analytic:

$$\mathbf{B}(x, y, z) = \begin{pmatrix} \frac{2zx - (a/b)Fy}{s} \\ \frac{2zy + (b/a)Fx}{s} \\ 1 - s \end{pmatrix}. \quad (1)$$

The flux function and pressure are

$$\psi(x, y, z) = \frac{x^2 + y^2 + 4z^2 + |\mathbf{B}(x, y, z)|^2 - 2 + \epsilon^2}{4}, \quad p = p_a - 2\psi. \quad (2)$$

Although nonnegativity is not obvious from this formula, below we will show that ψ is exactly a sum of two squares.

Choose

$$0 < \delta < \frac{(1 - \epsilon)^2}{4}, \quad (3)$$

and restrict the field to the closed solid torus

$$\Omega_\delta = \{\mathbf{x} \in \mathcal{U}_\epsilon : 0 \leq \psi(\mathbf{x}) \leq \delta\}. \quad (4)$$

The field is analytic on an open neighborhood of this entire solid torus, including its magnetic axis and boundary.

For a completely specified example, take

$$\epsilon = \frac{1}{4}, \quad \delta = \frac{1}{64}, \quad p_a = 1. \quad (5)$$

2. Explicit field lines and global coordinates

The verification becomes simple in coordinates adapted to the field lines.

Let u, v satisfy

$$u^2 + v^2 < \frac{1}{4},$$

and let t be an angle modulo 2π . Set

$$q^2 = u^2 + v^2, \quad L = \sqrt{\frac{1 + \sqrt{1 - 4q^2}}{2}}.$$

Define the position map

$$\begin{aligned} x(u, v, t) &= a \left[L \cos t + \frac{u \cos t + v \sin t}{L} \right], \\ y(u, v, t) &= b \left[L \sin t + \frac{v \cos t - u \sin t}{L} \right], \\ z(u, v, t) &= v \cos 2t - u \sin 2t. \end{aligned} \quad (6)$$

Write this map as $\mathbf{r}(u, v, t)$.

Two central identities are

$$\mathbf{B}(\mathbf{r}(u, v, t)) = \frac{\partial \mathbf{r}}{\partial t} \quad (7)$$

and

$$\det \frac{\partial(x, y, z)}{\partial(u, v, t)} = -ab = -\sqrt{1 - \epsilon^2}. \quad (8)$$

Consequently, the field lines are precisely the curves with u, v fixed.

Here t is a field-line parameter, **not physical time**, and generally it is not identical to the cylindrical toroidal angle.

The map is globally one-to-one

It is useful to verify that (6) is a genuine coordinate system, not merely a local parametrization.

Define

$$w = X + iY, \quad \eta = \frac{s-1}{2} + iz.$$

Equation (6) implies

$$\eta = (u + iv)e^{-2it}, \quad w = e^{it} \left(L + \frac{\eta}{L} \right). \quad (9)$$

Conversely, from any point of $\mathcal{U}_\epsilon$, recover

$$q = |\eta| < \frac{1}{2}, \quad L = \sqrt{\frac{1 + \sqrt{1 - 4|\eta|^2}}{2}},$$

and then

$$e^{it} = \frac{w}{L + \eta/L}, \quad u + iv = \eta e^{2it}. \quad (10)$$

The denominator cannot vanish because $q < L^2$. The first ratio has unit modulus, since

$$\left| L + \frac{\eta}{L} \right|^2 = 1 + 2 \operatorname{Re} \eta = s = |w|^2.$$

Thus the inverse is unique, with t interpreted modulo 2π .

This proves that (6) is an analytic diffeomorphism

$$\left\{ u^2 + v^2 < \frac{1}{4} \right\} \times S^1 \longrightarrow \mathcal{U}_\epsilon. \quad (11)$$

Solenoidality

Since the contravariant components of $\mathbf{B}$ in these coordinates are $(0, 0, 1)$, and the volume Jacobian is constant,

$$\nabla \cdot \mathbf{B} = \frac{1}{ab} \frac{\partial}{\partial t}(ab) = 0. \quad (12)$$

For another way to see why this works, before the constant stretching of the x and y directions, the field has cylindrical components

$$B_r = \frac{2z}{r}, \quad B_\varphi = \frac{\sqrt{2r^2 - r^4 - 4z^2}}{r}, \quad B_z = 1 - r^2.$$

Its divergence vanishes immediately. The constant affine pushforward used in (1) preserves divergence. Force balance, which does **not** generally survive such a deformation, will be checked separately.

3. The flux surfaces, magnetic axis, and non-axisymmetry

Introduce the auxiliary scalar

$$H = \frac{x^2 + y^2 + 4z^2}{2} + \frac{|\mathbf{B}|^2}{2}. \quad (13)$$

Using (6)–(7), the three oscillator energies are

$$\begin{aligned} \frac{x^2 + (\partial_t x)^2}{2} &= \frac{a^2}{2}(1 + 2u), \\ \frac{y^2 + (\partial_t y)^2}{2} &= \frac{b^2}{2}(1 - 2u), \\ \frac{4z^2 + (\partial_t z)^2}{2} &= 2(u^2 + v^2). \end{aligned} \quad (14)$$

Adding them gives

$$H = 1 + 2\epsilon u + 2(u^2 + v^2) = 1 - \frac{\epsilon^2}{2} + 2 \left[\left(u + \frac{\epsilon}{2} \right)^2 + v^2 \right]. \quad (15)$$

Comparing with (2),

$$\boxed{\psi(\mathbf{r}(u, v, t)) = \left(u + \frac{\epsilon}{2} \right)^2 + v^2.} \quad (16)$$

This proves both

$$\psi \geq 0 \quad \text{and} \quad \mathbf{B} \cdot \nabla \psi = 0. \quad (17)$$

For completeness, the labels themselves have explicit single-valued Cartesian expressions:

$$\boxed{u = \frac{1}{2} \left[X^2 + \left(\frac{B_x}{a} \right)^2 - 1 \right], \quad v = \frac{1}{2} \left[XY + \frac{B_x B_y}{ab} \right].} \quad (18)$$

Thus (16), together with (18), also gives a manifestly nonnegative alternative to the Cartesian formula (2).

Nested tori

In the label plane, Ω_δ corresponds to the disk

$$D_\delta = \left\{ (u, v) : \left(u + \frac{\epsilon}{2} \right)^2 + v^2 \leq \delta \right\}.$$

Condition (3) guarantees that this disk lies strictly inside $u^2 + v^2 < 1/4$, because

$$\sqrt{u^2 + v^2} \leq \frac{\epsilon}{2} + \sqrt{\delta} < \frac{1}{2}. \quad (19)$$

Therefore

$$\Omega_\delta = \mathbf{r}(D_\delta \times S^1)$$

is a compact embedded solid torus.

For any $0 < c \leq \delta$, its flux surface is explicitly

$$\mathbf{r} \left(-\frac{\epsilon}{2} + \sqrt{c} \cos \beta, \sqrt{c} \sin \beta, t \right), \quad (\beta, t) \in S^1 \times S^1. \quad (20)$$

These are smooth, closed, bounded, embedded, nested tori. Moreover,

$$\nabla \psi \neq 0 \quad \text{everywhere off the magnetic axis.} \quad (21)$$

The magnetic axis

The set $\psi = 0$ is the single field line

$$u = -\frac{\epsilon}{2}, \quad v = 0.$$

Equation (6) simplifies on this line to

$$\gamma_\epsilon(t) = \left(\sqrt{1 - \epsilon^2} \cos t, \sqrt{1 - \epsilon^2} \sin t, \frac{\epsilon}{2} \sin 2t \right). \quad (22)$$

In physical cylindrical coordinates this is

$$R_{\text{axis}} = \sqrt{1 - \epsilon^2}, \quad Z_{\text{axis}}(\phi) = \frac{\epsilon}{2} \sin 2\phi. \quad (23)$$

For the concrete choice $\epsilon = 1/4$,

$$R_{\text{axis}} = \frac{\sqrt{15}}{4}, \quad Z_{\text{axis}}(\phi) = \frac{1}{8} \sin 2\phi. \quad (24)$$

Why the field is genuinely non-axisymmetric

For $\epsilon \neq 0$, this axis is nonplanar: no plane can contain a curve whose Cartesian components have the independent Fourier dependences $\cos t, \sin t, \sin 2t$.

Also, because $p = p_a - 2\psi$, the axis is precisely the set where $\nabla p = 0$, equivalently where $(\nabla \times \mathbf{B}) \times \mathbf{B} = 0$. Any continuous Euclidean rotational symmetry of the field would therefore have to preserve this curve.

A smooth embedded closed curve invariant under a nontrivial continuous group of Euclidean rotations must be a circle. Our axis is not even planar. Hence:

$$\boxed{\text{The field is not axisymmetric about any straight line in } \mathbb{R}^3.} \quad (25)$$

This is not an axisymmetric equilibrium viewed in translated or tilted coordinates.

4. Exact scalar-pressure force balance

The useful standard identity is the stationary Euler–MHD correspondence: if the field's convective derivative is a gradient, the Lorentz force is obtained by subtracting the magnetic-energy gradient. This is the same algebra underlying the Bernoulli formulation of stationary Euler flow. [London Mathema...](#)

For the present construction, every x and y component in (6) has frequency 1, while z has frequency 2. Consequently,

$$\frac{\partial^2 \mathbf{r}}{\partial t^2} = (-x, -y, -4z). \quad (26)$$

Since $\partial_t \mathbf{r} = \mathbf{B}(\mathbf{r})$, this is the exact identity

$$\boxed{(\mathbf{B} \cdot \nabla) \mathbf{B} = (-x, -y, -4z) = -\nabla Q, \quad Q = \frac{x^2 + y^2 + 4z^2}{2}.} \quad (27)$$

Using

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = (\mathbf{B} \cdot \nabla) \mathbf{B} - \nabla \frac{|\mathbf{B}|^2}{2},$$

we obtain

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = -\nabla \left(Q + \frac{|\mathbf{B}|^2}{2} \right) = -\nabla H. \quad (28)$$

But (15)–(16) give

$$H = 1 - \frac{\epsilon^2}{2} + 2\psi.$$

Therefore

$$\boxed{(\nabla \times \mathbf{B}) \times \mathbf{B} = -2\nabla\psi = \nabla p.} \quad (29)$$

This is exact throughout the solid torus, including the axis by analyticity. The scalar Q is only an auxiliary quantity in the proof; it is **not** an additional physical force in the equilibrium.

The pressure gradient is nonzero everywhere off the axis. Choosing $p_a > 2\delta$ makes the pressure positive throughout the domain.

There is also a nonzero axial current. Direct evaluation gives

$$\boxed{(\nabla \times \mathbf{B})(\gamma_\epsilon(t)) = \frac{4}{1 - \epsilon^2} \mathbf{B}(\gamma_\epsilon(t)).} \quad (30)$$

5. Rotational transform: exactly -2

To distinguish genuine winding from a coordinate convention, use physical cylindrical coordinates and define the geometric meridional angle by

$$\theta_{\text{geom}} = \arg \left[(R - R_{\text{axis}}) + i(z - Z_{\text{axis}}(\phi)) \right]. \quad (31)$$

Its continuous lift along a field line counts poloidal turns about the axis in the (R, Z) sections.

First, the toroidal angle increases strictly:

$$\boxed{\mathbf{B} \cdot \nabla \phi = \frac{abF}{x^2 + y^2} > 0.} \quad (32)$$

At fixed u, v , the xy projection of (6) is a positively traversed ellipse surrounding the origin. Hence $t \mapsto t + 2\pi$ corresponds to exactly one physical toroidal turn:

$$\Delta\phi = 2\pi. \quad (33)$$

Compute the poloidal winding at $\epsilon = 0$

The formulas extend smoothly to $\epsilon = 0$. Write

$$u + iv = qe^{i\alpha}.$$

Then the magnetic axis is $R = 1, z = 0$, and (9) becomes

$$\boxed{\frac{R^2 - 1}{2} + iz = qe^{i(\alpha - 2t)}.} \quad (34)$$

Therefore the meridional displacement makes two negative turns when t increases by 2π .

Replacing $(R^2 - 1)/2$ by $R - 1$ does not change the winding, since their ratio is the positive function $(R + 1)/2$. Thus

$$\Delta\theta_{\text{geom}} = -4\pi, \quad \iota = \frac{\Delta\theta_{\text{geom}}}{\Delta\phi} = -2. \quad (35)$$

Continue to $\epsilon > 0$

For fixed $c > 0$ and β , continuously increase ϵ while following the field line

$$u = -\frac{\epsilon}{2} + \sqrt{c} \cos \beta, \quad v = \sqrt{c} \sin \beta.$$

The embedding stays nonsingular, the line remains disjoint from the magnetic axis, and toroidal advance stays positive. Its meridional winding is an integer that cannot change during this deformation.

Consequently, throughout the allowed family,

$$\boxed{\iota = -2} \quad (36)$$

with the orientation convention (31). Reversing the poloidal-angle convention changes the sign, not the nonzero magnitude.

However, every component of (6) is 2π -periodic in t , so

$$\boxed{\mathbf{r}(u, v, t + 2\pi) = \mathbf{r}(u, v, t).} \quad (37)$$

Thus all field lines close after one toroidal turn, after winding twice poloidally, and the one-turn Poincaré map is exactly the identity.

6. Reproducible implementation and checks

I prepared a NumPy implementation containing the Cartesian field, flux function, pressure, field-line embedding, magnetic axis, and numerical differentiation checks:

[`</>` Download the Python implementation and verification script](#)

Run it with

`</>` Bash

```
python exact_nonaxisymmetric_mhd_integer_transform.py
```



For $\epsilon = 1/4$, $\delta = 1/64$, and 300 random points in the solid torus, the computed maximum residuals were approximately

$$\begin{aligned} |\nabla \cdot \mathbf{B}| &< 5.6 \times 10^{-16}, \\ |(\nabla \times \mathbf{B}) \times \mathbf{B} - \nabla p| &< 4.2 \times 10^{-15}, \\ |\mathbf{B} \cdot \nabla \psi| &< 1.2 \times 10^{-16}. \end{aligned}$$

The geometric field-line check gave one toroidal turn, minus two poloidal turns, and a return distance of approximately 2.2×10^{-16} . These computations corroborate, rather than replace, the analytic verification above.

The result is an exact non-axisymmetric, nonconstant-pressure equilibrium with a complete nested toroidal foliation and nonzero integer transform. The limitation is specifically the return map: this construction does not provide noninteger transform or a nonidentity one-turn return.

`</>` exact_nonaxisymmetric_mhd_integer_transform.py
Code

... Sources